

Unsupervised Learning - Study Notes

What is unsupervised learning?

Unsupervised learning works with data that has no labels. The model tries to discover structure on its own. The main tasks are clustering, dimensionality reduction and anomaly detection.

k-means clustering

k-means partitions the data into k clusters. Each cluster is represented by a centroid. The algorithm repeatedly assigns every point to its nearest centroid and then recomputes the centroids until the assignments stop changing. The number of clusters k must be chosen in advance. The result is sensitive to the initial centroids, and k-means++ initialization helps. k-means also assumes clusters that are roughly spherical and of similar size.

Choosing k

The elbow method plots the within-cluster sum of squares (also called inertia) against k. The best k is the point where the curve bends like an elbow and the improvement slows down sharply. The silhouette score measures how similar a point is to its own cluster compared with the other clusters. It ranges from -1 to 1, and higher values are better.

Hierarchical clustering

Agglomerative hierarchical clustering starts with every point as its own cluster and repeatedly merges the two closest clusters. The result is a tree called a dendrogram. Cutting the dendrogram at a chosen height gives the final clusters, so the number of clusters does not have to be fixed in advance.

DBSCAN

DBSCAN is a density-based clustering algorithm. It groups points that are closely packed together and labels points in low-density regions as noise or outliers. Unlike k-means, DBSCAN does not need the number of clusters in advance and it can find clusters of arbitrary shape. Its two main parameters are eps, the neighborhood radius, and min_samples, the minimum number of points needed to form a dense region.

Dimensionality reduction

Principal Component Analysis (PCA) projects the data onto new axes called principal components, which point in the directions of maximum variance. The explained variance ratio shows how much information each component keeps. A common rule is to keep enough components to explain 95 percent of the variance. t-SNE and UMAP are nonlinear methods used mainly to visualize high-dimensional data in two or three dimensions.

Anomaly detection

Anomaly detection finds rare observations that differ from the majority of the data. Isolation Forest is a popular method: it isolates points with random splits, and anomalies need fewer splits to be isolated. Typical uses are fraud detection and equipment monitoring.